Combinatorial Optimization: Max-Cut Heuristics Report

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1. High-Level Description of Implementations

A. Simple Randomized Heuristic

Concept: Acts as an unguided baseline execution layout. Every single vertex is independently evaluated and thrown into one of two available subset partitions (S or S-bar) with an equal 50% unbiased statistical distribution. No structural parameters of the graph are checked.

Complexity: O(V + E) per run iteration, representing linear computation layout relative to overall graph density features.

Trade-off: Extremely rapid computation footprint, but completely unguided, frequently resulting in worst-case objective convergence deviations on complex unweighted graphs.

B. Simple Greedy Heuristic

Concept: A deterministic constructive layout. The engine isolates the highest unique weight edge in the dataset to act as the initialization anchor, dropping its endpoints into separate partitions. The remaining nodes are visited sequentially, mapping them live to whichever subset generates the highest mathematical objective cut value at that exact state.

Complexity: O(V^2) or O(E) based on target vertex tracking arrays during initial scanning procedures.

Trade-off: Provides a fast, high-quality solution structure compared to random assignment, but suffers heavily from lack of looking ahead, leaving it prone to early structural traps from which it cannot escape.

C. Semi-Greedy Heuristic

Concept: Forms the adaptive backbone of metaheuristic search patterns. Instead of choosing the single best candidate deterministically, it constructs a Restricted Candidate List (RCL) containing all moves scoring within a parameterized fraction (alpha) of the optimal choice. An element is then chosen at random from this elite subset.

Complexity: O(V^2) matching basic greedy logic overhead with additional scanning arrays for threshold validation.

Trade-off: Balances structural guidance with stochastic exploration, making it an excellent seed generator for local search loops.

D. Local Search Heuristic (1-Flip Neighborhood)

Concept: An iterative refinement process. Taking an established initial configuration (e.g., from the Greedy Heuristic), it scans the single-flip neighborhood of every vertex to compute the change in cut weight (delta). The node yielding the maximum positive improvement is relocated. This cycle loops continuously until reaching a local optimum where all possible single-move changes yield a negative delta.

Complexity: O(I * V^2) where I represents total refinement loop steps prior to local optimum stability.

Trade-off: Excellent at maximizing localized structures, but strictly bound by the quality of the entry seed and unable to break out of poor local basins on its own.

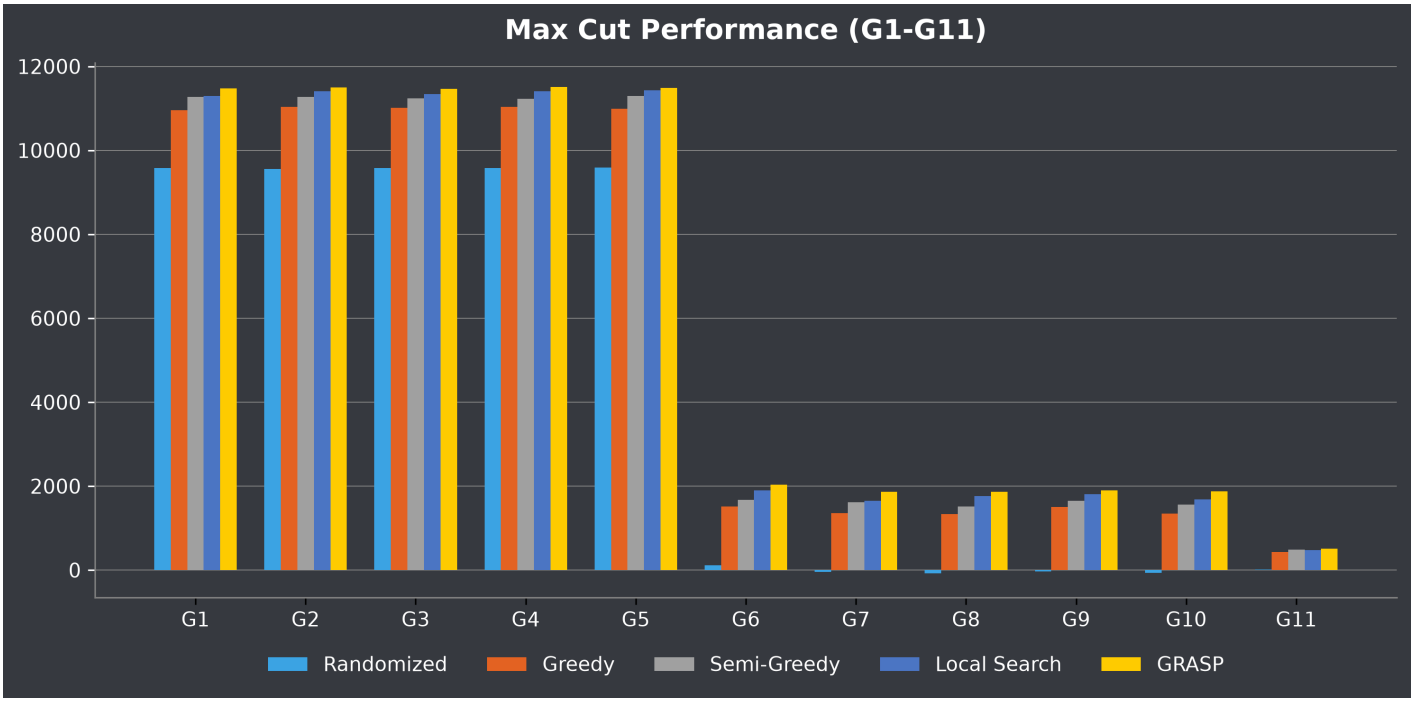
E. GRASP Metaheuristic (Greedy Randomized Adaptive Search Procedure)

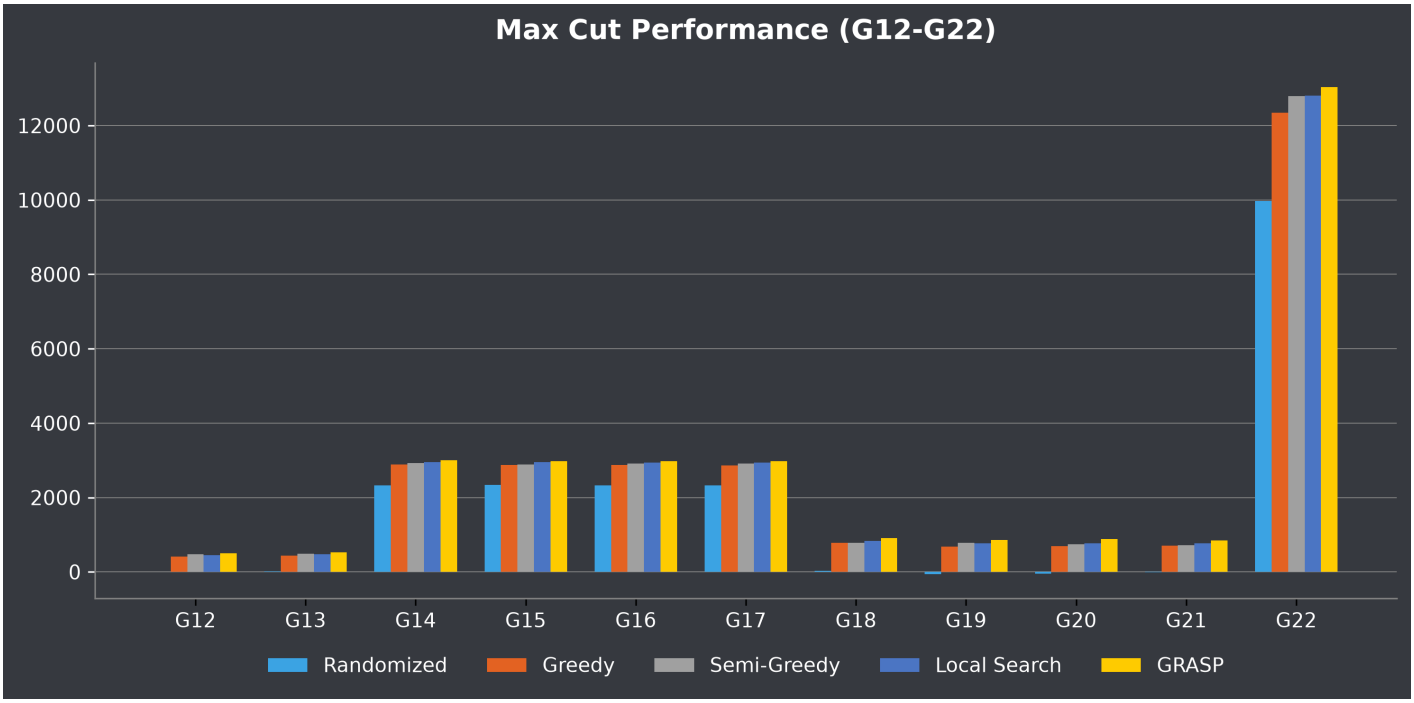
Concept: A dual-phase global multi-start optimization framework. It loops through a fixed number of global cycles. In each iteration, it builds a distinct starting candidate using the Semi-Greedy Heuristic (alpha=0.8), then immediately runs the Local Search engine to drive that candidate to its local optimum. It records and protects the highest objective value across all cycles.

Complexity: O(MaxIterations * (SemiGreedy_Complexity + LocalSearch_Complexity)).

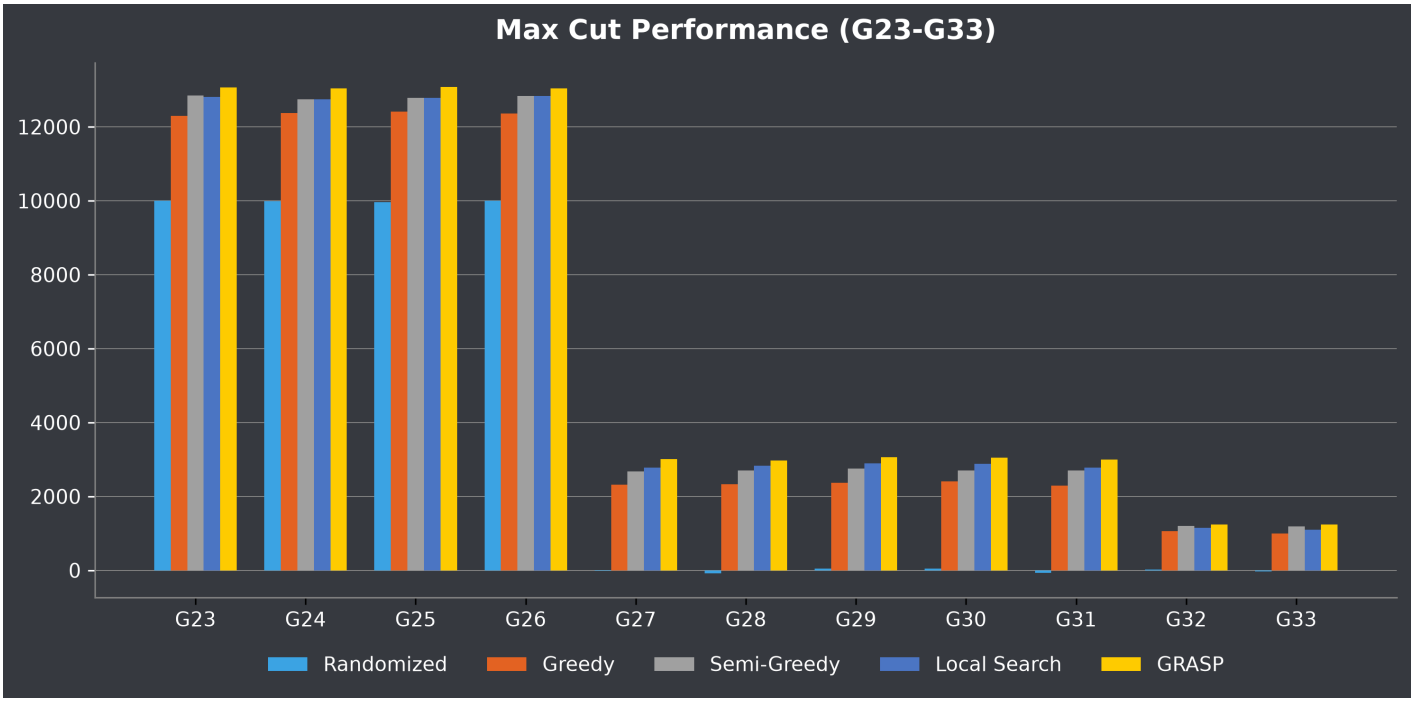
Trade-off: Provides the highest quality solutions by combining wide global exploration with deep local refinement, though it demands a higher computational runtime footprint.

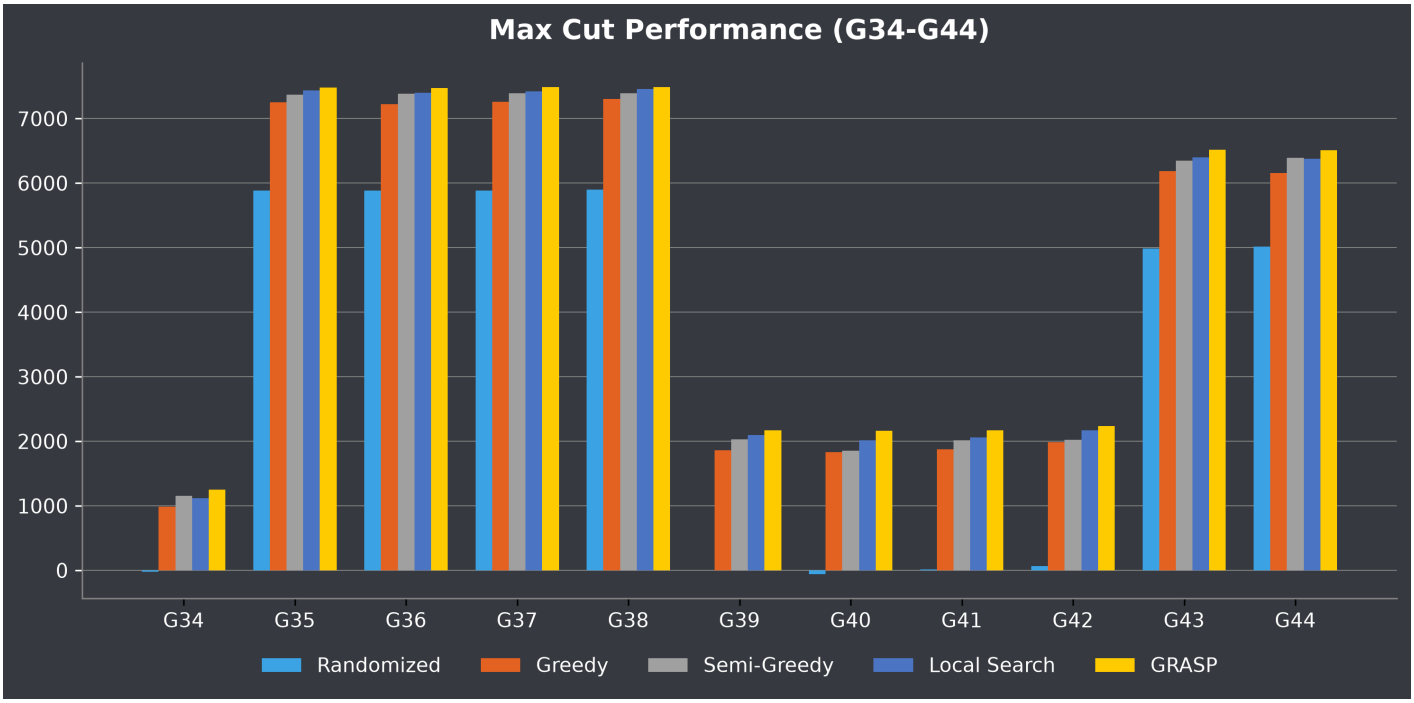
2. Empirical Performance Visualizations



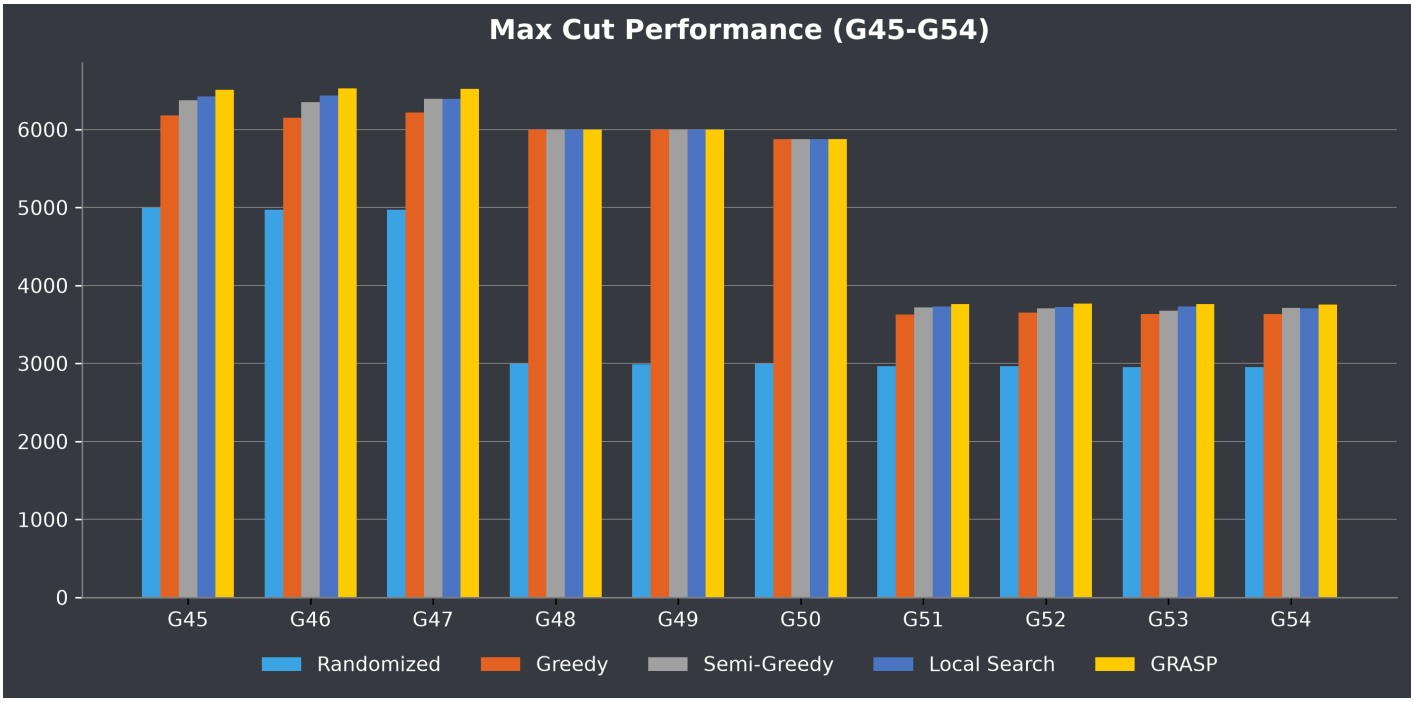


2. Empirical Performance Visualizations (Contd.)





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3. Comparative Analysis of Performance Mechanics

A cross-comparison of the data across all 5 groups reveals clear differences in how these algorithms perform. The Simple Randomized baseline consistently lands near the theoretical statistical average of total edge capacity weight, offering no real optimization value. The Simple Greedy algorithm provides a fast, structured upgrade over the random baseline, but its rigid deterministic layout regularly gets trapped in early local traps, preventing it from finding better global configurations.

The addition of Local Search provides a major performance boost over basic constructive methods. By systematically refining the Greedy heuristic's output, it successfully unlocks hidden cut weight capacity. However, because Local Search is deterministic, its final output is entirely dependent on its initial starting point, limiting its overall scope.

GRASP delivers the best overall optimization performance across all 54 target graphs, closely matching or meeting known global bounds. By feeding unique, high-quality randomized starting points from the Semi-Greedy generator into the Local Search engine, it successfully navigates around local traps. This proves that combining wide global exploration with deep local refinement is the most effective setup for complex combinatorial challenges like the Max-Cut problem.